

Talabat Operations dashboard

Software Requirement Specification

Version 1.0

Prepared by: Ahad Salim ALjabri
21/5/2026



Table of content

1. Introduction	3
1.1 Purpose	3
1.2 Scope	3
1.3 Overview	3
2. General Description	3
2.1 Users	3
2.2 Objectives	3
3. Data Requirements	3
4. Functional Requirements	3
5. Non-Functional Requirements	4
6. Tools & Technologies	4
7. Expected Outputs (Include Prototype)	4
8. Constraints	4
9. Project Timeline (Use clickup/ Notion/ etc)	5
10. Financial Requirements	5
10.1 Cost Breakdown	5
10. 2 Payment Terms	5
11. Key Insights	5
12. Recommendations	5



1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) documents the design, development, and analytical outcomes of the Talabat Operations Dashboard, a business intelligence solution built using Microsoft Power BI. The project transforms 100,000 raw food delivery order records into actionable insights across six analytical domains: executive overview, delivery performance, revenue analysis, restaurant and menu analysis, operational intelligence, and customer behaviour.

1.2 Scope

The dashboard covers the complete Talabat order lifecycle across 7 Egyptian cities — Cairo, Giza, Alexandria, Mansoura, Tanta, Zagazig, and Assiut — over a 15-day operational period. It is designed to serve multiple stakeholder audiences including executives, operations managers, finance teams, restaurant partners, logistics coordinators, and marketing analysts.

1.3 Overview

Dataset

100,000 Total Orders	9,000 Unique Customers	1,000 Unique Restaurants
500 Unique Drivers	7 Cities Covered	15 Days Data Period
EGP 26,892,573 Total Revenue	EGP 268.93 Avg Order Value	85.2% Delivery Success

2. General Description

2.1 Users

1. Executive Management
2. Operations Manager
3. Finance Team
4. Restaurant Partners

2.2 Objectives

The primary objectives of this dashboard are:

- Transform 100,000 raw Talabat order records into clear, actionable business insights
- Provide a 6-page focused dashboard where each page addresses one specific business question
- Identify and resolve real data quality issues in the source dataset
- Enable data-driven decision making for operations, revenue, and customer strategy
- Demonstrate proficiency in Power BI DAX, and visual design



3. Data Requirements

3.1 Dataset Source

The dataset is `talabat_enhanced_orders` — a single flat fact table containing 100,000 food delivery order records from Talabat Egypt. The data covers 7 cities over a 15-day operational period, from Kaggle.

3.2 Column Reference

Column Name	Type	Description
Order_ID	Text	Unique order identifier
User_ID	Text	Customer identifier — 9,000 unique values
Restaurant_ID	Text	Restaurant identifier — 1,000 unique values
Driver_ID	Text	Driver identifier — 500 unique values
Item_Name	Text	Menu item ordered — 9 categories
Quantity	Integer	Number of items per order
Total_Price	Decimal	Order total in Egyptian Pounds (EGP)
Order_Date	Date	Date of order placement
Order_Time	Time	Time of order placement
Delivery_Duration_Minutes	Integer	Actual delivery time in minutes
Delivery_Distance_km	Decimal	Distance from restaurant to customer
Delivery_Delay	Text	Actual duration minus 37-min benchmark. Stored as text — converted via <code>VALUE()</code>
City	Text	One of 7 Egyptian cities
Payment_Method	Text	Cash / Credit Card / Wallet
Order_Status	Text	Delivered / Cancelled / In Transit
Traffic_Level	Text	Low / Medium / High
Driver_Vehicle	Text	Motorbike / Car / Bicycle
Day_of_Week	Text	Empty in source — fixed via calculated column <code>FORMAT([Order_Date], dddd)</code>
order-hour	Integer	Hour of day (0-23)
Customer_Lat	Decimal	Customer latitude for map visual
Customer_Lon	Decimal	Customer longitude for map visual



4. Functional Requirements

4.1 Dashboard Pages

Page	Title	Purpose & Key Visuals
1	Executive Overview	4 cards, line chart (orders over time), bar chart (orders by city), 2 donuts (order status + payment method)
2	Delivery Performance	4 cards, bar chart (delivery time by city), clustered bar (traffic impact), scatter (distance vs time), stacked bar (on-time vs delayed)
3	Revenue & Orders	4 cards, bar chart (top items), column charts (revenue by hour + day), gauge (orders per customer)
4	Restaurant & Menu	3 cards, bar chart (top 10 restaurants), donut (orders by item), leaderboard table, clustered bar (items by city)
5	Operations & Traffic	3 cards, map visual, clustered bar (traffic vs delay), donut (vehicle split), matrix (vehicle vs traffic)
6	Customer Behaviour	4 cards, column charts (orders by day + hour), stacked bar (payment by city), bar chart (avg order value by city)

4.2 DAX Measures — Used in Dashboard

All 25 measures below are active and used in at least one visual across the 6 dashboard pages. Only functions actually present in the DAX expressions are listed.

Measure Name	DAX Expression	Function Category
Total Orders	COUNTROWS(talabat_enhanced_orders)	Aggregation — COUNTROWS
Total Revenue	SUM(talabat_enhanced_orders[Total_Price])	Aggregation — SUM
Avg Order Value	AVERAGE(talabat_enhanced_orders[Total_Price])	Aggregation — AVERAGE
Avg Delivery Time (mins)	AVERAGE(talabat_enhanced_orders[Delivery_Duration_Minutes])	Aggregation — AVERAGE
Avg Delivery Distance (km)	AVERAGE(talabat_enhanced_orders[Delivery_Distance_km])	Aggregation — AVERAGE
Unique Customers	DISTINCTCOUNT(talabat_enhanced_orders[User_ID])	Aggregation — DISTINCTCOUNT
Unique Restaurants	DISTINCTCOUNT(talabat_enhanced_orders[Restaurant_ID])	Aggregation — DISTINCTCOUNT
Unique Drivers	DISTINCTCOUNT(talabat_enhanced_orders[Driver_ID])	Aggregation — DISTINCTCOUNT
Delivered Orders	CALCULATE(COUNTROWS(...), Order_Status = "Delivered")	Filter — CALCULATE
Cancelled Orders	CALCULATE(COUNTROWS(...), Order_Status = "Cancelled")	Filter — CALCULATE



Delayed Orders	CALCULATE(COUNTROWS(...), VALUE(Delivery_Delay) > 37)	Filter — CALCULATE + Information — VALUE
On Time Orders	CALCULATE(COUNTROWS(...), VALUE(Delivery_Delay) <= 37)	Filter — CALCULATE + Information — VALUE
Avg Delay (mins)	AVERAGEX(FILTER(..., VALUE(Delay) > 0), VALUE(Delay))	Aggregation — AVERAGEX + Filter — FILTER + Information — VALUE
Avg Delivery Time High Traffic	CALCULATE([Avg Delivery Time (mins)], Traffic_Level="High")	Filter — CALCULATE
Avg Delivery Time Low Traffic	CALCULATE([Avg Delivery Time (mins)], Traffic_Level="Low")	Filter — CALCULATE
High Traffic Rate	DIVIDE(CALCULATE([Total Orders], Traffic_Level="High"), [Total Orders], 0)	Math — DIVIDE + Filter — CALCULATE
Delivery Success Rate	DIVIDE([Delivered Orders], [Total Orders], 0)	Math & Statistical — DIVIDE
Cancellation Rate	DIVIDE([Cancelled Orders], [Total Orders], 0)	Math & Statistical — DIVIDE
Delay Rate	DIVIDE([Delayed Orders], [Total Orders], 0)	Math & Statistical — DIVIDE
Orders per Customer	DIVIDE([Total Orders], [Unique Customers], 0)	Math & Statistical — DIVIDE
Revenue per Customer	DIVIDE([Total Revenue], [Unique Customers], 0)	Math & Statistical — DIVIDE
Revenue per Order	DIVIDE([Total Revenue], [Total Orders], 0)	Math & Statistical — DIVIDE
Revenue per Restaurant	DIVIDE([Total Revenue], [Unique Restaurants], 0)	Math & Statistical — DIVIDE
Data Period (Days)	DATEDIFF(MIN([Order_Date]), MAX([Order_Date]), DAY)	Date & Time — DATEDIFF, MIN, MAX
Orders per Customer Target	15 (constant)	Constant — Gauge target value

5. Non-Functional Requirements

- Performance: Dashboard loads within 5 seconds; all 25 measures calculate without timeout on 100,000 rows
- Usability: Each page has one clear purpose.
- Accuracy: All measures use DIVIDE() to prevent division-by-zero errors
- Data Integrity: VALUE() applied to Delivery_Delay column to resolve text/numeric type mismatch.
- Maintainability: All measures organized into 7 display folders in the Power BI data model.
- Accessibility: All values use human-readable format strings.



6. Tools & Technologies

Tool / Technology	Version / Type	Purpose
Microsoft Power BI Desktop	Latest (2026)	Dashboard development, data modeling, DAX measures
DAX (Data Analysis Expressions)	Power BI built-in	28 calculated measures across 7 function categories
Power Query	Built-in	Data loading and initial transformation
Azure Maps / ArcGIS	Built-in visual	Geographic map on Operations page (Page 5)
Dataset: talabat_enhanced_orders	Kaggle	100,000 order records — single fact table

7. Expected Outputs

The project delivers a single Power BI .pbix file containing:

- 6 fully built report pages, each focused on one analytical domain
- Data quality fixes for 1 identified issues (Day_of_Week empty column)



8. Constraints

- Static dataset — no live data connection; dashboard reflects snapshot data only
- Day_of_Week source column is empty — requires calculated column workaround using `FORMAT([Order_Date], "dddd")`
- Driver_Availability column excluded from analysis — offline drivers showed completed deliveries, indicating unreliable data
- No time intelligence due to dataset covering only 15 days
- Map visual requires Azure Maps or ArcGIS license for full geographic rendering
- Currency displayed as EGP — original dataset had no currency label; EGP applied based on Egyptian city data

9. Project Timeline

Phase	Activity	Duration
1 — Data Exploration	Load dataset, audit all 21 columns, identify data types and quality issues	Day 1
2 — SRS Documentation	Complete this SRS document	Day 1-2
3 — Data Quality Fixes	Fix Delivery_Delay (VALUE()), create Day_of_Week_Fixed column, reassess delay benchmark	Day 1-2
4 — Data Modeling	Create 25 DAX measures across 7 display folders	Day 2-3
5 — Dashboard Build	Build 6 report pages with all visuals, cards, and slicers	Day 3-5
6 — Review & Fix	Fix visuals	Day 5-6
7 — Currency Update	Show all revenue format strings to EGP	Day 6 -7

10. Financial Requirements

10.1 Cost Breakdown

⇒ Sometimes no need to breakdown just general cost and that's it

Item	COST	Description
Dashboard Development	150 OMR	Dashboard Development
Data Cleaning	60 OMR	Data Cleaning
DAX Measures & KPIs	80 OMR	DAX Measures & KPIs
Interactive Visuals	100 OMR	Interactive Visuals



Deployment & Publishing	40 OMR	Deployment & Publishing
Documentation	40 OMR	Documentation
Total	470 OMR	

10. 2 Payment Terms

- 50% upfront
- 50% after completion

11. Key Insights

#	Insight	Data Evidence
1	Sunday is the busiest day — 30% above weekdays	Sunday: 17,317 orders vs avg 13,300 on weekdays
2	Talabat is a true 24/7 platform — no peak hours	Every hour: 4,100 to 4,260 orders, flat distribution
3	Customers are highly loyal — 11.1 orders each	9,000 customers / 100,000 orders = 11.1 repeat rate
4	All 3 payment methods equally used	Cash 33.5%, Card 33.2%, Wallet 33.3%
5	Shawarma leads revenue but margin is small	Shawarma EGP 3.06M vs Sushi EGP 2.93M — gap is only 4%
6	All 7 cities within 3% of each other in orders	Range: 14,044 (Mansoura) to 14,538 (Zagazig)
7	Traffic has minimal impact on delivery time	High vs Low traffic: difference less than 0.2 minutes
8	Vehicle type has no performance difference	Motorbike, Car, Bicycle all average 37.5 mins
9	9.8% cancellation rate is worth investigating	9,812 orders cancelled — highest revenue risk

12. Recommendations

- Increase driver availability on Sundays — 30% higher demand justifies temporary surge staffing
- Investigate cancellation reasons — 9.8% cancellation (9,812 orders) is the single biggest revenue risk; target reduction below 5%
- Review Zagazig operations — highest delay rate (50.9%) and slowest average delivery time (37.67 mins) across all cities
- Maintain all three payment methods equally — 33% split shows no preference; removing any option would alienate one third of customers
- Consider Shawarma as a featured item in promotional campaigns — top revenue item with broad appeal across all cities
- Connect to a live data source — current dashboard is a static snapshot; live refresh would make it operational rather than analytical

